[P7] Seven: no family of eighteen cards, so $19 \le g(7) \le 21$

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Abstract

No pairwise intersecting 7-uniform family of eighteen cards has transversal number seven. With the 21-card family of [P5, §5] this gives $19 \le g(7) \le 21$. The argument deletes the cards of a symbol of degree four and splits into two branches. In one, the twelve-card picture of [P3, §2a] reappears on fourteen cards, and every candidate kernel turns out to contains four disjoint triples, and four disjoint triples give a cover of size five. In the other, a second deletion leaves a ten-card core, and no core exists. Both finite steps were computed twice, by separately written programs, and the programs reproduce the counts of [P3] at $r = 6$.

The argument is elementary counting plus computation; it does not use the proposed proofs of [P5]. Nothing here has been reviewed by a human referee or a proof assistant. No priority is claimed. See the note at the end.

1. Setting

Let H be a pairwise intersecting family of eighteen cards, each a set of seven symbols, with $\tau(H) = 7$. We show that no such H exists.

Theorem 1 (computer-assisted). *There is no pairwise intersecting 7-uniform family of eighteen cards with transversal number seven. Hence $g(7) \ge 19$, and with [P5, §5], $19 \le g(7) \le 21$.*

As in [P3], it helps to read the cards as points and the symbols as blocks: a symbol is the set of cards that contain it, and its size is its degree.

2. Degree four

Lemma 1. *The maximum degree of H is exactly four.*

Proof. If every degree were at most three, a card would meet the other seventeen through seven symbols reaching at most two cards each, and $17 \le 14$ is false.

Suppose a symbol x has degree $d \ge 5$. Delete its cards and call the remaining $m = 18 - d$ cards F. A cover of F together with x covers H, so $\tau(F) \ge 6$. We show $\tau(F) \le 5$.

- $m \le 10$: pair the cards; each pair shares a symbol.
- $m = 11$: a symbol meeting three cards, with a pairing of the other eight, gives five. If every symbol meets at most two cards, the 55 pairs need 55 different symbols, but the 77 incidences allow at most 38.
- $m = 12$ or 13: a symbol meeting four or more cards leaves at most nine, and those have a cover of size four (pair them if there are fewer than nine; for nine, argue as for eleven, with 36 pairs against at most 31 symbols), so the total is at most five. Otherwise every symbol meets at most three cards of F, and Lemma 2 gives three disjoint triples. They meet nine cards; the other three or four pair off through two more symbols, five in all. □

Lemma 2 (computer-assisted). *Let F be twelve or thirteen pairwise intersecting cards of size seven in which every symbol meets at most three cards. Then the symbols meeting three cards contain three pairwise disjoint triples.*

How it was checked. As in Lemma 6 below, each card p needs room: the triples at p, plus the cards sharing no triple with p (each needs its own symbol of size two), number at most seven. Removing repeated copies of a triple keeps this condition and does not change which triples are disjoint, so it is enough to consider sets of triples. If the largest number of disjoint triples is k, fix k disjoint ones (all such placements are equivalent); no other triple avoids them all. A SAT model shows that $k = 1$ and $k = 2$ are impossible for twelve and for thirteen cards, with two solvers (CaDiCaL, Glucose) and two cardinality encodings. As controls, the same model is satisfiable with $k = 3$ (twelve and thirteen cards) and $k = 4$ (twelve).

3. The first deletion

Fix a symbol x of degree four and let G be the other fourteen cards. A cover of G together with x covers H, so $\tau(G) \geq 6$. Every symbol has degree at most four in G. Either G has a symbol of degree four (§5) or it does not (§4).

4. Branch A: the kernel has maximum degree three

4.1 The three profiles

Each card of G meets the other thirteen through its seven symbols, and each symbol reaches at most two of them. Writing (t, b, s) for the numbers of its symbols of degree three, two and one in G, we need $t + b + s = 7$ and $2t + b \geq 13$. So every card has profile $(7, 0, 0)$ or $(6, 1, 0)$, and there are no symbols of degree one.

This is the situation of [P3, §2a] with fourteen cards in place of twelve. A $(6, 1, 0)$ card meets every other card exactly once, and a $(7, 0, 0)$ card meets exactly one other card twice. So the degree-two symbols form a matching F on the c cards of type $(6, 1, 0)$, the doubled pairs form a matching E on the others, and the degree-three symbols are the triangles of a decomposition of the pair multigraph

$$M = K_{14} + E - F.$$

The number of triangles is $(98 - c)/3$, and c is even, so $c \in \{2, 8, 14\}$. A triangle holds at most one edge of E, so no triangle occurs twice.

4.2 The obstruction

Lemma 3. *If $\tau(G) \geq 6$, the triangles of G contain no four that are pairwise disjoint.*

Proof. Four disjoint triangles meet twelve cards through four symbols. The two cards left meet each other, through a fifth symbol. That is a cover of size five. □

Lemma 4 (computer-assisted). *For each $c \in \{2, 8, 14\}$, every triangle decomposition of M contains four pairwise disjoint triangles.*

How it was checked. Let ν be the largest number of pairwise disjoint triangles.

- $\nu \leq 2$ is impossible for all three profiles, by a plain exhaustive search (13,614,095, 26,427,987 and 193,839,399 search nodes for $c = 14, 8, 2$).

- $\nu = 3$. Fix three disjoint triangles. The five cards outside them hold no triangle, since that would be a fourth. The placements of the three triangles fall into orbits under the automorphisms of M, which permute the pairs of F and of E and swap the two cards inside a pair. There are 266,560, 367,696 and 505,120 placements, in 6, 122 and 60 orbits for $c = 14, 8, 2$. From one placement in each orbit, the rest of the decomposition is searched to exhaustion. None completes.

The $\nu = 3$ step was done twice, with no code in common. The first route takes the orbits from nauty and completes each one with a search tree. The second takes the orbits by union-find under explicit generators of the automorphism group, and completes each one with a SAT solver, forbidding any four disjoint triangles that appear. Both give the same orbit counts, and neither finds a decomposition. For $c = 14$ a third check, a plain exhaustive search of the whole profile (600,808,145 nodes), also finds none.

By Lemmas 3 and 4, $\tau(G) \leq 5$ in every case, so branch A is empty.

5. Branch B: a second symbol of degree four

Let y have degree four in G. Its cards are not cards of x. Delete them as well, and let K be the remaining ten cards, the **core**.

Lemma 5. *In the core:*

1. *every symbol meets at most three cards;*
2. *any two symbols of size three meet;*
3. *every card lies in a symbol of size three;*
4. *for every card p, the number of size-three symbols at p, counted with repetition, plus the number of cards that share no size-three symbol with p, is at most seven.*

Proof.

1. A symbol meeting four cards of K: delete them; the six cards left pair off, and each pair shares a symbol. With x, y and that symbol this is a cover of size six.
2. Two disjoint triples meet six cards through two symbols; the other four cards pair off through two more; with x and y that is six.
3. A card meets nine others through seven symbols. With no symbol of size three, each reaches at most one, and $7 < 9$.
4. Each card that shares no triple with p needs its own symbol of size two with p. These symbols, and the triples at p, are all different symbols of p, which has seven. □

Lemma 6 (computer-assisted). *No multiset of 3-subsets of ten points satisfies 2, 3 and 4 of Lemma 5.*

How it was checked. Twice. A SAT model of conditions 2 to 4, with each triple allowed up to seven times (condition 4 bounds the repetition by seven), is unsatisfiable. A separate enumeration of intersecting triple systems, using the bound of three repetitions (a card holding a triple m times reaches two cards through it, and $2 + 2(7 - m) \geq 9$), finds none that leaves room in every card. The same enumeration program, run at $r = 6$ on eight points, returns the 463 cores of [P3, §3].

So branch B is empty, and Theorem 1 is proved. □

6. Calibration

Each program was run at $r = 6$ before it was used at $r = 7$, where the answers are known:

program	at $r = 6$	known value
exhaustive kernel search, profiles 12 and 6	69,120 and 558,672	[P3, §2a]
kernel search with a fixed matching, both routes	finds kernels	they exist
core enumeration, eight points	463 cores	[P3, §3]
core SAT model, eight points	satisfiable	cores exist

7. Verification

claim	script
Lemma 2, twelve and thirteen cards	`code/r7_small_families.py`
Lemma 4, $\nu \leq 2$; plain search of $c = 14$	`code/r7_kernel_exhaustive.c`
Lemma 4, $\nu = 3$, first route	`code/r7_kernel_matched.c` (needs nauty)
Lemma 4, $\nu = 3$, second route	`code/r7_kernel_sat.py`
Lemma 6, enumeration	`code/r7_cores.c` (needs nauty)
Lemma 6, SAT	`code/r7_cores_sat.py`
all of the above with expected answers	`code/verify_g7_lower.py`
the recorded runs	`results/g7_lower_checks.txt`
the upper bound $g(7) \leq 21$	`code/verify_witness_r7_21.py`

8. What is and is not established

- Nothing from [P5] is used. An earlier version of this paper excluded degree five and more through [P5, Theorem 1]; Lemma 2 now does it directly.
- The arguments of Lemma 1, §3, §4.1, Lemma 3 and Lemma 5 are short and elementary; they have been read, not reviewed.
- Lemmas 2, 4 and 6 are computations. Each was done by two programs with no code in common, and each program reproduces known counts at $r = 6$. They carry no proof certificates.
- Whether $g(7) \geq 19$ or $g(7) \leq 21$ is already known has not been checked beyond a short search.

This work was prepared with AI assistance (ChatGPT, OpenAI; Claude, Anthropic), used for algebraic derivation, for drafting and rewriting code and text, for running and re-running the computations, and for auditing the papers against their own scripts. The same tools were used in the review, so the review carries the same caveat. All statements were checked by the author, who is responsible for them. No priority is claimed and the result has not been independently reviewed by a human or a proof assistant; the repository README sets out the division of labour and the open obligations in full.

References

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